RICCARDO SALZMANN

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EXPERIENCE

SMT / SportsMEDIA Technology [Durham, NC] *March of 2025 – Present*
Software Engineer

- Owned architecture and delivery of a multithreaded C++ application that ingested external API data to render kick trajectories; results utilized live, on-air, during NBC television broadcast of Super Bowl LX.
- Developed video fingerprints to identify on-air cameras across NBA television broadcasts; accelerated DCT hashing with CUDA; handled buffer synchronization, as well as out-of-order frames, for reliable graphics selection.
- Developed confidence-scored sketch recognition to convert freehand annotations into geometric shapes with animated transitions; results utilized live, on-air, for telestration during NBC television broadcasts of Sunday Night Football.
- Owned automated player-data / video synchronization across hockey, football, and basketball using Ceres optimization and Hungarian matching; authored documentation and worked directly with operators to troubleshoot calibration.
- Led a NASCAR telemetry application from stakeholder interviews through installer delivery; analyzed 6M+ records in 2D / 3D views and post-race reports, uncovering issues in Vector hardware devices.
- Built a shared C++ / OpenGL line-rendering library for 6+ applications, supporting configurable tessellation, textures, and miter, bevel, and round joins across NBA, NFL, and NHL television broadcasts.
- Delivered putting rings and distance overlays aired during the Golf Channel television broadcast of the Masters Tournament; created a CAD course model, used fiducials to calibrate the 3D mesh, and projected selections onto terrain.

Quantum America (Real Estate Startup) [Austin, TX] *March of 2026 – Present*
Freelance Software Engineer [Remote]

- Architected and built an interactive 3D web application for exploring architectural spaces using TypeScript and React; translated client requirements into application design and delivered the implementation.
- Transformed architectural models into interactive 3D home interiors and exteriors for the web using Three.js.
- Automated Blender-to-web asset builds with Node.js and Python, adding validation, Draco compression, versioned exports, and catalog rollback.
- Reverse-engineered a vendor Android app to enable PC control of a Linux e-ink display; built a Python/JavaScript interface for image uploads, conversion, and scheduled playback.

EDAG Group [Troy, MI / Ingolstadt, Germany] *June of 2022 – March of 2025*
Software Engineer Intern

- Built a C++/Qt dashboard simulator that decoded live CAN bus signals and replayed recorded vehicle data from configurable DBC files, enabling signal validation through a QML display.
- Cut firmware updates from 60 to 5 seconds with a Qt/C++ updater for a commercial RV manufacturer.

OPEN SOURCE & LEADERSHIP

Rust Standard Library & Compiler Tests *Nov. of 2024 – March of 2025*

- Modularized C-compatible primitive types for FFI testing (PR #136334, merged); prototyped cross-target tests comparing Rust type sizes and signedness against Clang (PR #133944, unmerged).

Head Coach, Junior Varsity Soccer Team | East Chapel Hill High School *[Chapel Hill, NC]* *Aug. of 2025 – Oct. of 2025*

- Managed a high school Junior Varsity soccer team through full season; planned and coached all training / games.

EDUCATION

University of Michigan [Ann Arbor, MI] *Degree Awarded in Dec. of 2024*
Bachelor of Science in Computer Science

TECHNICAL / PRACTICAL SKILLS

Computer Languages:	C++, C, Rust, Python, TypeScript, JavaScript, SQL, QML
Libraries / Tools:	Codex, Claude Code, CUDA, OpenGL, Qt, React, Three.js, Blender, SVN, Git, CMake, Linux
Engineering:	Multithreading, real-time rendering, computational geometry, CAN bus, embedded Linux, System Design
Languages:	English (native), German (fluent)